

RESTAURANT RATINGS & CONSUMER PREFERENCE ANALYSIS

Executive Summary

Understanding customer preferences is critical for restaurant success. Restaurants that align their offerings with customer expectations are more likely to achieve higher satisfaction ratings, stronger customer loyalty, and sustainable profitability.

This project analyzes restaurant ratings, consumer demographics, cuisine preferences, and behavioral patterns to identify factors influencing restaurant performance and customer satisfaction.

The analysis revealed significant differences in customer ratings across restaurants, occupations, and preference categories. Several restaurants consistently outperformed competitors, while consumer demographics and preferences played a substantial role in determining overall satisfaction.

Business Problem

Restaurant owners and investors need data-driven insights to understand customer behavior and make informed business decisions.

This project aims to answer the following questions:

- Which restaurants receive the highest ratings?
 - Which customer segments provide the highest ratings?
 - How do consumer preferences influence satisfaction?
 - Which cuisines are most attractive to customers?
 - What opportunities exist for restaurant investment and growth?
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Dataset Overview

Dataset Components

The project combines multiple datasets containing:

- Restaurant Information
- Consumer Information
- Consumer Preferences
- Restaurant Cuisines
- Customer Ratings

Dataset Size

- Restaurants: 130
 - Consumers: 120
 - Ratings: 1,000+
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Data Modeling

A star-schema model was implemented to support efficient analysis.

Fact Table

- Fact_Ratings

Dimension Tables

- Dim_Consumers
- Dim_Restaurants
- Dim_Restaurant_Cuisines
- Dim_Consumer_Preferences

Relationships were established between the fact table and all dimension tables to enable multidimensional analysis.

Data Cleaning & Preparation

The following preprocessing activities were completed:

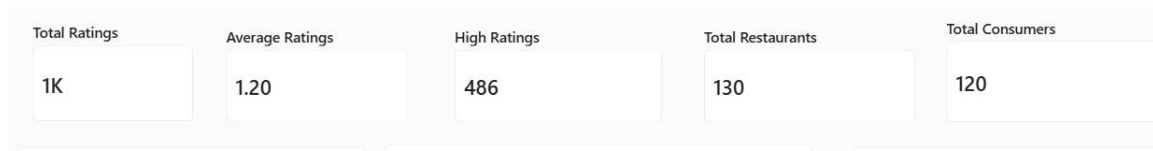
Data Cleaning

- Handled missing values
- Replaced blanks with "Unknown" where appropriate
- Standardized text categories
- Corrected data types

Data Modeling

- Built star schema
 - Validated relationships
 - Optimized model structure
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Dashboard KPIs

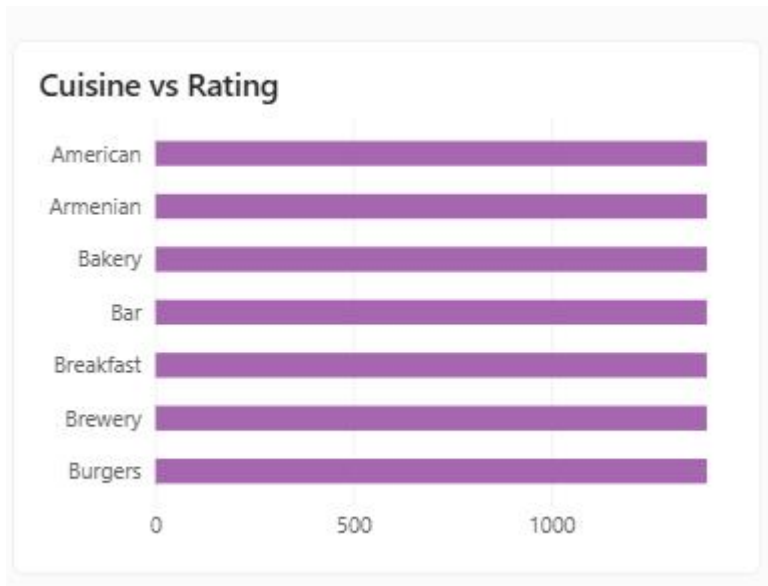


KPIs:

- Total Ratings: 1,000+
 - Average Rating: 1.20
 - High Ratings: 486
 - Total Restaurants: 130
 - Total Consumers: 120
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Analysis & Findings

Cuisine Performance Analysis



Findings

Certain cuisine categories consistently received higher ratings than others.

This indicates stronger customer demand and greater customer satisfaction within specific cuisine segments.

Business Implication

Cuisine selection plays a critical role in restaurant success.

Top-Rated Restaurants



Findings

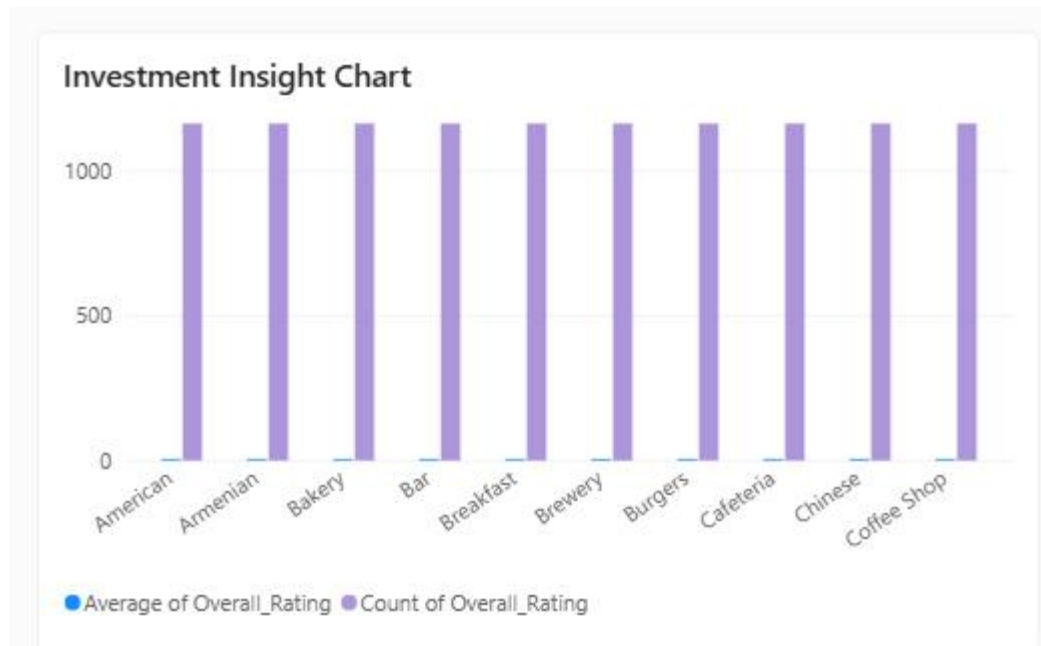
Several restaurants consistently achieved superior ratings compared to competitors.

These restaurants demonstrate strong alignment between customer expectations and service delivery.

Business Implication

Their practices may serve as benchmarks for other restaurants.

Occupation vs Rating Analysis



Findings

Employed consumers generally provided higher ratings than students and unemployed consumers.

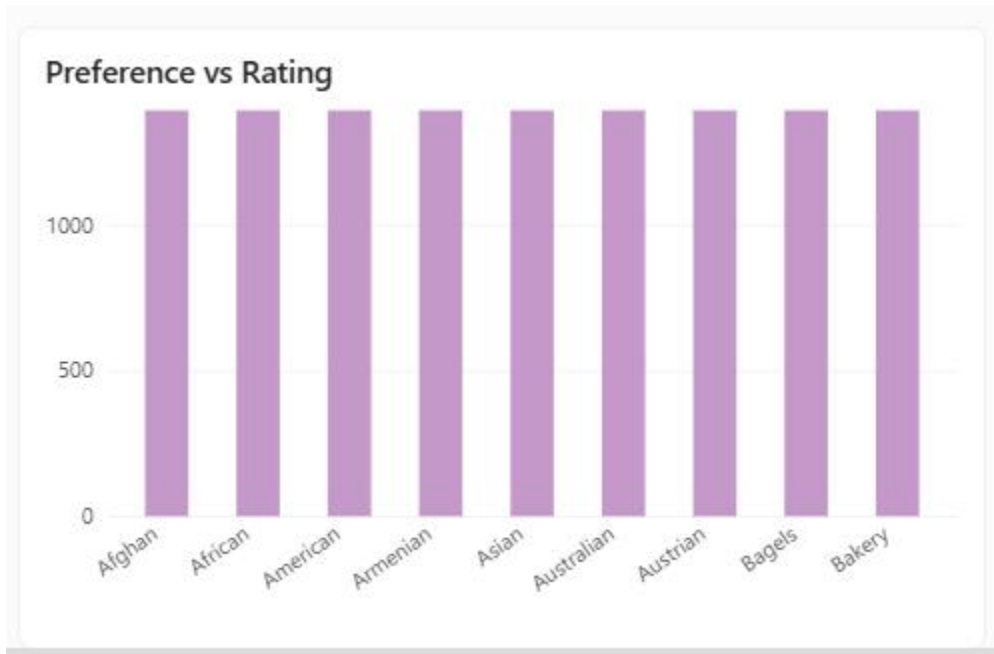
Possible Explanations

- Higher purchasing power
- Different dining expectations
- Greater appreciation of service quality

Business Implication

Working professionals represent an attractive target market.

Consumer Preference Analysis



Findings

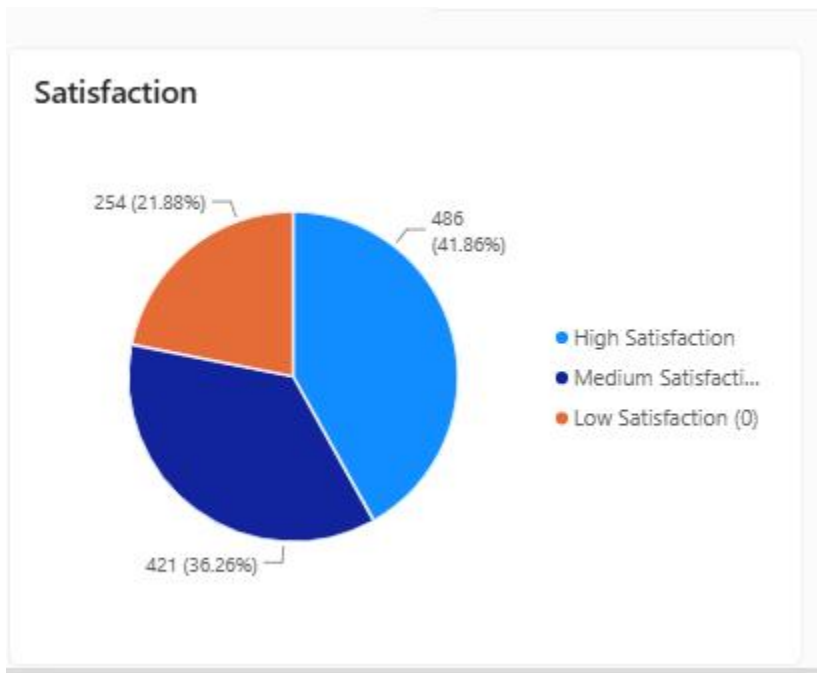
Customer preferences strongly influence satisfaction outcomes.

Consumers tend to reward restaurants that align with their preferred cuisine and dining experiences.

Business Implication

Understanding customer preferences is essential for menu development and marketing.

Satisfaction Distribution



Findings

High satisfaction ratings represent the largest segment of customer feedback. This indicates that many restaurants successfully meet customer expectations.

Key Insights

Insight 1

Consumer preferences significantly impact restaurant ratings.

Insight 2

Some cuisine categories consistently outperform others.

Insight 3

Top-performing restaurants demonstrate strong customer alignment.

Insight 4

Occupation influences customer rating behavior.

Insight 5

Customer satisfaction varies considerably across restaurants.

Insight 6

High-rated restaurants provide valuable lessons for underperforming competitors.

Recommendations

Business Growth Recommendations

1. Invest in High-Performing Cuisine Categories

Focus expansion efforts on cuisine types with consistently high ratings.

Expected Outcome:

Higher customer demand and stronger profitability.

2. Target Working Professionals

Develop promotions and dining experiences tailored to employed consumers.

Expected Outcome:

Higher spending and increased customer retention.

3. Improve Underperforming Restaurants

Actions:

- Review service quality
- Improve menu offerings
- Enhance customer experience

Expected Outcome:

Higher ratings and improved competitiveness.

4. Use Consumer Preference Data

Actions:

- Personalize marketing campaigns
- Adjust menus based on customer demand
- Optimize restaurant positioning

Expected Outcome:

Stronger customer engagement.

5. Benchmark Top-Rated Restaurants

Actions:

- Analyze successful restaurant practices
- Replicate effective strategies

Expected Outcome:

Improved performance across locations.

6. Expand Popular Cuisine Offerings

Actions:

- Increase availability of highly rated cuisines
- Introduce complementary menu options

Expected Outcome:

Greater market share and customer satisfaction.

Conclusion

The analysis demonstrates that customer preferences, cuisine selection, and demographic characteristics significantly influence restaurant ratings.

Businesses that align their offerings with customer expectations are better positioned to achieve higher satisfaction levels, stronger loyalty, and long-term success.

Full Dashboard Screenshot

